

5. Ethics and Safety Statement

With the advancement in the reasoning capabilities of DeepSeek-R1, we deeply recognize the potential ethical risks. For example, R1 can be subject to jailbreak attacks, leading to the generation of dangerous content such as explosive manufacturing plans, while the enhanced reasoning capabilities enable the model to provide plans with better operational feasibility and executability. Besides, a public model is also vulnerable to further fine-tuning that could compromise inherent safety protections.

In Supplementary D.3, we present a comprehensive safety report from multiple perspectives, including performance on open-source and in-house safety evaluation benchmarks, and safety levels across multiple languages and against jailbreak attacks. These comprehensive safety analyses conclude that the inherent safety level of the DeepSeek-R1 model, compared to other state-of-the-art models, is generally at a moderate level (comparable to GPT-4o (2024-05-13)). Besides, when coupled with the risk control system, the model’s safety level is elevated to a superior standard.

6. Conclusion, Limitation, and Future Work

We present DeepSeek-R1-Zero and DeepSeek-R1, which rely on large-scale RL to incentivize model reasoning behaviors. Our results demonstrate that pre-trained checkpoints inherently possess substantial potential for complex reasoning tasks. We believe that the key to unlocking this potential lies not in large-scale human annotation but in the provision of hard reasoning questions, a reliable verifier, and sufficient computational resources for reinforcement learning. Sophisticated reasoning behaviors, such as self-verification and reflection, appeared to emerge organically during the reinforcement learning process.

Even if DeepSeek-R1 achieves frontier results on reasoning benchmarks, it still faces several capability limitations, as outlined below:

Structure Output and Tool Use: Currently, the structural output capabilities of DeepSeek-R1 remain suboptimal compared to existing models. Moreover, DeepSeek-R1 cannot leverage tools, such as search engines and calculators, to improve the performance of output. However, as it is not hard to build an RL environment for structure output and tool use, we believe the issue will be addressed in the next version.

Token efficiency: Unlike conventional test-time computation scaling approaches, such as majority voting or Monte Carlo Tree Search (MCTS), DeepSeek-R1 dynamically allocates computational resources during inference according to the complexity of the problem at hand. Specifically, it uses fewer tokens to solve simple tasks, while generating more tokens for complex tasks. Nevertheless, there remains room for further optimization in terms of token efficiency, as instances of excessive reasoning—manifested as overthinking—are still observed in response to simpler questions.

Language Mixing: DeepSeek-R1 is currently optimized for Chinese and English, which may result in language mixing issues when handling queries in other languages. For instance, DeepSeek-R1 might use English for reasoning and responses, even if the query is in a language other than English or Chinese. We aim to address this limitation in future updates. The limitation may be related to the base checkpoint, DeepSeek-V3-Base, mainly utilizes Chinese and English, so that it can achieve better results with the two languages in reasoning.

Prompting Engineering: When evaluating DeepSeek-R1, we observe that it is sensitive to

prompts. Few-shot prompting consistently degrades its performance. Therefore, we recommend users directly describe the problem and specify the output format using a zero-shot setting for optimal results.

Software Engineering Tasks: Due to the long evaluation times, which impact the efficiency of the RL process, large-scale RL has not been applied extensively in software engineering tasks. As a result, DeepSeek-R1 has not demonstrated a huge improvement over DeepSeek-V3 on software engineering benchmarks. Future versions will address this by implementing rejection sampling on software engineering data or incorporating asynchronous evaluations during the RL process to improve efficiency.

Beyond specific capability limitations, the pure RL methodology itself also presents inherent challenges:

Reward Hacking: The success of pure RL depends on reliable reward signals. In this study, we ensure reward reliability through a reasoning-domain rule-based reward model (RM). However, such dependable RMs are difficult to construct for certain tasks, such as writing. If the reward signal is assigned by a model instead of predefined rules, it becomes more susceptible to exploitation as training progresses, which means the policy model may find shortcuts to hack the reward model. Consequently, for complex tasks that cannot be effectively evaluated by a reliable reward model, scaling up pure RL methods remains an open challenge.

In this work, for tasks that cannot obtain a reliable signal, DeepSeek-R1 uses human annotation to create supervised data, and only conduct RL for hundreds of steps. We hope in the future, a robust reward model can be obtained to address such issues.

With the advent of pure RL methods like DeepSeek-R1, the future holds immense potential for solving any task that can be effectively evaluated by a verifier, regardless of its complexity for humans. Machines equipped with such advanced RL techniques are poised to surpass human capabilities in these domains, driven by their ability to optimize performance iteratively through trial and error. However, challenges remain for tasks where constructing a reliable reward model is inherently difficult. In such cases, the lack of a robust feedback mechanism may hinder progress, suggesting that future research should focus on developing innovative approaches to define and refine reward structures for these complex, less verifiable problems.

Furthermore, leveraging tools during the reasoning process holds significant promise. Whether it’s utilizing tools like compilers or search engines to retrieve or compute necessary information, or employing external tools—such as biological or chemical reagents, to validate final results in the real world, this integration of tool-augmented reasoning could dramatically enhance the scope and accuracy of machine-driven solutions.

7. Author List

The list of authors is organized by contribution role, with individuals listed alphabetically by their first name within each category. Authors marked with an asterisk (*) are no longer affiliated with our team.

Core Contributors: Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, Xiao Bi, Xiaokang Zhang, Xingkai Yu, Yu Wu, Z.F. Wu, Zhibin Gou, Zhihong Shao, Zhuoshu Li, Ziyi Gao,

Contributions of the Core Authors: Peiyi Wang and Daya Guo jointly demonstrated that outcome-based RL induces the emergence of long Chain-of-Thought patterns in LLMs, achieving

breakthrough reasoning capabilities. They contributed equally to the creation of R1-Zero, and their work laid the foundation for R1. Daya Guo also contributed to the RL training stability of MOE models. Junxiao Song proposed the GRPO algorithm, implemented the initial version, and introduced rule-based rewards for math tasks. The GRPO algorithm was subsequently refined by Peiyi Wang and Runxin Xu. Zhibin Gou proposed a large PPO clipping strategy to enhance GRPO performance, demonstrating its significance alongside Zhihong Shao and Junxiao Song. Regarding data iteration, reward design, and evaluation, specific teams led efforts across different domains: Qihao Zhu, Z.F. Wu, and Dejian Yang focused on code tasks; Zhihong Shao, Zhibin Gou, and Junxiao Song focused on math tasks; and Peiyi Wang, Ruoyu Zhang, Runxin Xu, and Yu Wu led efforts for other reasoning and general tasks. Additionally, Qihao Zhu and Zhihong Shao contributed to the data selection strategy for RL training, while Zhuoshu Li and Yu Wu co-led the data labeling efforts for the entire project. On the system side, Xiao Bi, Xingkai Yu, Shirong Ma, Xiaokang Zhang, Haowei Zhang, and Ziyi Gao implemented the RL pipeline, optimizing system efficiency and addressing stability issues in large-scale training. Finally, Zhibin Gou, Daya Guo, and Ruoyu Zhang oversaw the final training phase and monitored the model training dynamics. Zhibin Gou led the development of the R1-distill series.

Contributors: Aixin Liu, Bing Xue, Bingxuan Wang, Bochao Wu, Bei Feng, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chong Ruan, Damai Dai, Deli Chen, Dongjie Ji, Erhang Li, Fangyun Lin, Fucong Dai, Fuli Luo*, Guangbo Hao, Guanting Chen, Guowei Li, H. Zhang, Hanwei Xu, Honghui Ding, Huazuo Gao, Hui Qu, Hui Li, Jianzhong Guo, Jiashi Li, Jingchang Chen, Jingyang Yuan, Jinhao Tu, Junjie Qiu, Junlong Li, J.L. Cai, Jiaqi Ni, Jian Liang, Jin Chen, Kai Dong, Kai Hu*, Kaichao You, Kaige Gao, Kang Guan, Kexin Huang, Kuai Yu, Lean Wang, Lecong Zhang, Liang Zhao, Litong Wang, Liyue Zhang, Lei Xu, Leyi Xia, Mingchuan Zhang, Minghua Zhang, Minghui Tang, Mingxu Zhou, Meng Li, Miaojun Wang, Mingming Li, Ning Tian, Panpan Huang, Peng Zhang, Qiancheng Wang, Qinyu Chen, Qiushi Du, Ruiqi Ge, Ruisong Zhang, Ruizhe Pan, Runji Wang, R.J. Chen, R.L. Jin, Ruyi Chen, Shanghao Lu, Shangyan Zhou, Shanhuang Chen, Shengfeng Ye, Shiyu Wang, Shuiping Yu, Shunfeng Zhou, Shuting Pan, S.S. Li, Shuang Zhou, Shaoqing Wu, Shengfeng Ye, Tao Yun, Tian Pei, Tianyu Sun, T. Wang, Wangding Zeng, Wen Liu, Wenfeng Liang, Wenjun Gao, Wenqin Yu*, Wentao Zhang, W.L. Xiao, Wei An, Xiaodong Liu, Xiaohan Wang, Xiaokang Chen, Xiaotao Nie, Xin Cheng, Xin Liu, Xin Xie, Xingchao Liu, Xinyu Yang, Xinyuan Li, Xuecheng Su, Xuheng Lin, X.Q. Li, Xiangyue Jin, Xiaojin Shen, Xiaosha Chen, Xiaowen Sun, Xiaoxiang Wang, Xinnan Song, Xinyi Zhou, Xianzu Wang, Xinxia Shan, Y.K. Li, Y.Q. Wang, Y.X. Wei, Yang Zhang, Yanhong Xu, Yao Li, Yao Zhao, Yaofeng Sun, Yaohui Wang, Yi Yu, Yichao Zhang, Yifan Shi, Yiliang Xiong, Ying He, Yishi Piao, Yisong Wang, Yixuan Tan, Yiyang Ma*, Yiyuan Liu, Yongqiang Guo, Yuan Ou, Yudian Wang, Yue Gong, Yuheng Zou, Yujia He, Yunfan Xiong, Yuxiang Luo, Yuxiang You, Yuxuan Liu, Yuyang Zhou, Y.X. Zhu, Yanping Huang, Yaohui Li, Yi Zheng, Yuchen Zhu, Yunxian Ma, Ying Tang, Yukun Zha, Yuting Yan, Z.Z. Ren, Zehui Ren, Zhangli Sha, Zhe Fu, Zhean Xu, Zhenda Xie, Zhengyan Zhang, Zhewen Hao, Zhicheng Ma, Zhigang Yan, Zhiyu Wu, Zihui Gu, Zijia Zhu, Zijun Liu*, Zilin Li, Ziwei Xie, Ziyang Song, Zizheng Pan, Zhen Huang, Zhipeng Xu, Zhongyu Zhang, Zhen Zhang,